

Transformer

Parameter	Requirement
Turns ratio	1:8 to 1:10 (step-up)
Output voltage	1.0 – 1.5 kV
Frequency range	100 Hz – 10 MHz
Secondary config	Center-tapped, 180° outputs
Load type	Capacitive, ~1 nF per output
Isolation voltage	≥ 2 kV

Commercial Transformer Research

No commercial part met the full specification. The fundamental obstacle is a three-way conflict between turns ratio, output voltage, and frequency range. RF wideband transformers achieve the frequency range but are limited to signal-level voltages (≤ 500 V) and current ratings under 250 mA. Audio transformers like Lundahl achieve the 1:20 turns ratio and high isolation voltage (3 kV) but roll off above 100 kHz. Power transformers like the Hammond 740 reach the correct voltage and turns ratio (1:20.8, 2.4 kV center-tapped) but operate only at 50/60 Hz line frequency because their silicon-steel cores saturate above a few hundred Hz. Pulse transformers approach the voltage and frequency targets but lack the required turns ratio and center-tap configuration.

Conclusion: Custom winding was the most viable path forward for the most ideal transformer that best fit our constraints.

Core Material/Geometry Selection

Toroid: Material 43 (NiZn ferrite) is electrically insulating and rated for impedance transformer use from 1–50 MHz, making it suitable for continuous RF operation.

U-Core: Material 77 (MnZn ferrite) offers higher permeability but is electrically conductive. At MHz frequencies, eddy currents within the core itself cause severe losses.

Toroids are preferred over U-core because the closed magnetic path minimizes leakage inductance, which directly determines the upper frequency limit when driving a capacitive load. Larger toroids are preferred because a bigger cross-sectional area (A_e) reduces flux density at low frequencies, preventing core saturation and allowing operation down to 100 Hz.

Turn Count Calculations

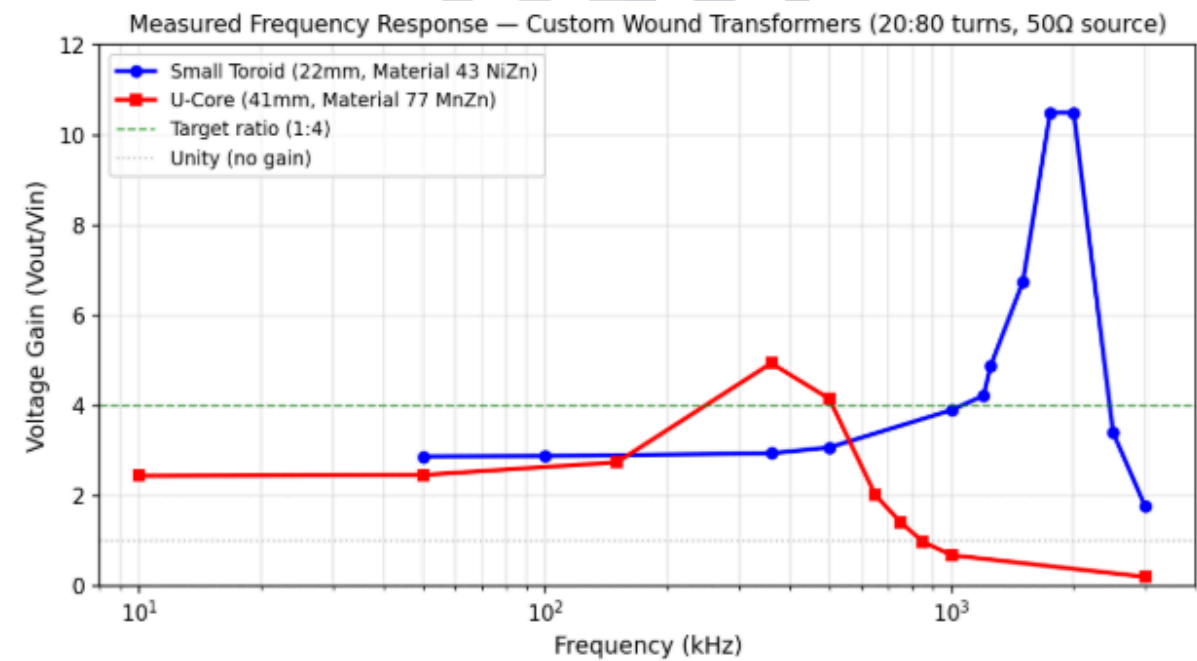
Minimum primary turns to avoid core saturation: $N_{min} = V / (4.44 \times f \times B_{max} \times Ae)$, evaluated at the lowest operating frequency and maximum primary voltage. Secondary turns are set by the 1:5 ratio per half (1:10 total center-tapped).

61 mm Material 43 Toroid: 65 primary turns, 650 secondary turns (325 each half)

Custom Winding Tests

Two cores were wound with 20:80 turns (1:4 ratio) and tested with a 50Ω source function generator swept from 10 kHz to 3 MHz.

Frequency	Small Toroid (Mat 43)	U-Core (Mat 77)	Observation
50 kHz	2.87x	2.46x	Both below 4x target
360 kHz	2.94x	4.94x (peak)	U-core peaks here only
1 MHz	3.90x	0.68x ↓	U-core dead; toroid works
1.75 MHz	10.5x (SRF)	—	Toroid self-resonance peak
3 MHz	1.76x	0.20x	Both heavily attenuated



Key findings: Material 43 NiZn maintains step-up gain past 1 MHz with SRF at ~1.75 MHz; Material 77 MnZn collapses to unity at 850 kHz and is effectively dead above

500 kHz. The consistent $\sim 2.9\times$ ratio below target (vs expected $4\times$) is caused by the 50Ω source impedance artifact, with the PA107 amplifier the ratio should approach $4\times$.

Next Steps

Wind FT-240-43 (61mm): 65 primary turns (AWG 22), 325 turns per secondary half
Theoretical lower cutoff ~ 86 Hz which is the best candidate for full operating range.

Continue small toroid testing: Retest with PA107 as low-impedance source, find exact SRF with open-circuit secondary, verify phase balance between center-tapped outputs.

Integrate with PA107: Run full test suite on wound FT-240-43, such as frequency response, phase balance, 1 nF capacitive load test, and incremental high voltage ramp to 1.5 kV.

